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Pooja Yogesh Athawale

AI/ML Engineer | Data Scientist | Generative AI Researcher

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Google Scholar: <https://scholar.google.com/citations?user=ZV7uDnkAAAAJ&hl=en>

Location: Phoenix, AZ

Professional Summary:

- AI/ML Engineer and Data Scientist with 6+ years of industry experience spanning data analytics, data engineering, statistical modeling, predictive modeling, machine learning, and NLP, complemented by graduate research in Generative AI, LLMs, RAG, model evaluation, and Trustworthy AI.
- Hands-on experience with LLMs, RAG, embeddings, vector databases, semantic search, Hugging Face Transformers, prompt engineering, LLM evaluation, confidence calibration, grounding, hallucination analysis, and response validation, with knowledge of Agentic AI, LLMops, and Model Context Protocol (MCP).
- Developed end-to-end ML and deep-learning solutions across classification, regression, forecasting, fraud detection, NLP, and computer vision using Python, SQL, Scikit-learn, XGBoost, TensorFlow, and PyTorch, with experience in cloud-based AI/ML and model deployment workflows.
- Graduate research in biomedical and trustworthy AI, including LLM/RAG reliability, breast-ultrasound representation learning, diabetic-retinopathy grading, uncertainty-aware modeling, evidence utilization, and human-in-the-loop (HITL) AI.

Research Publication & Recognition:

- **Google Scholar:** Publications: 2 peer-reviewed conference papers in Machine Learning & Biomedical AI | Google Scholar
- **AAS 2026:** Honorable Mention – Best Oral Presentation

Education:

- Master of Science in Data Science, Southern Arkansas University, USA
- Post Graduate Program in Cloud Computing, University of Texas at Austin, USA
- Bachelor's Degree in Information Technology, Dr. Babasaheb Ambedkar Marathwada University, INDIA.

Certifications:

Google AI Professional Certification | Claude 101 – Anthropic | AI Fluency Framework & Foundations – Anthropic | AI Fluency for Builders – Anthropic | AWS Educate Machine Learning Foundations | AWS Certified Cloud Practitioner | Salesforce Certified AI Associate | Tableau Desktop Specialist | Scrum Fundamentals Certified | Six Sigma Yellow Belt | Jira & Confluence

Technical Skills:

AI & Machine Learning: Machine Learning, Predictive Modeling, Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Feature Selection, Hyperparameter Tuning, Cross-Validation, Model Evaluation, Scikit-learn, XGBoost, TensorFlow, PyTorch

Generative AI & LLMs: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Engineering, LangChain, LangGraph, OpenAI API, Hugging Face Transformers, Sentence Transformers, Embeddings, Vector Databases, ChromaDB, Semantic Search, Grounding, Evidence Utilization, Structured Outputs, LLM Evaluation, Confidence Calibration, Hallucination & Failure Analysis, Responsible/Trustworthy AI

NLP, Deep Learning & Computer Vision: Natural Language Processing (NLP), TF-IDF, Word2Vec, BERT, Transformers, Sentiment Analysis, Text Classification, Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), ResNet, Variational Autoencoders (VAE), Representation Learning, Computer Vision

MLOps & AI Application Development: FastAPI, REST APIs, Docker, Kubernetes, MLflow, Model Deployment, Model Monitoring, Model Versioning, CI/CD, AWS SageMaker, Amazon Bedrock, Azure OpenAI

Programming & Data Analytics: Python, SQL, R, Pandas, NumPy, Statistical Analysis, Predictive Analytics, Regression Analysis, Forecasting, Data Cleaning, Data Transformation, Data Validation,

Exploratory Data Analysis (EDA)

Data Engineering, Databases & Cloud: ETL, Data Pipelines, Data Processing, Data Integration, Data Quality, Apache Spark, PySpark, SQL Server, PostgreSQL, Oracle, Data Modeling, Data Warehousing, AWS S3, EC2, Lambda, Glue, CloudWatch

Visualization & Tools: Tableau, Power BI, Excel, Git, Linux, Agile, Jira, UAT, Technical Documentation.

Professional Experience:

Research AI Engineer, Southern Arkansas University
AI/ML Research

Jan 2025 to Jul 2026

Responsibilities:

- Conducted graduate-level research in AI/ML, Generative AI, LLMs, RAG, Trustworthy AI, and Biomedical AI using Python, PyTorch, Scikit-learn, Hugging Face Transformers, and reproducible experimental workflows.
- Designed and implemented an end-to-end biomedical Retrieval-Augmented Generation (RAG) reliability framework across PubMedQA, MedQA, and MMLU-Med, integrating embeddings, semantic search, vector databases, prompt engineering, retrieval, LLM generation, and response evaluation.
- Developed embedding and retrieval pipelines using Sentence Transformers, BGE embeddings, ChromaDB, vector search, and cosine similarity, with fixed retrieval experiments to isolate and evaluate generation-side LLM behavior.
- Evaluated instruction-tuned LLMs including Qwen and Mistral using dataset-aware prompting and structured JSON outputs, measuring accuracy, confidence calibration, evidence utilization, grounding, hallucination behavior, and generation reliability.
- Designed LLM evaluation and response-validation workflows to identify overconfident incorrect predictions, reasoning failures, unsupported responses, invalid structured outputs, and cases where retrieved evidence was ignored or only partially utilized.
- Applied prompt engineering, context construction, structured-output generation, and response-validation techniques to improve consistency, grounding, and reliability of LLM-generated responses.
- Investigated LLM Ops and evaluation practices, including experiment tracking, prompt/model configuration management, reproducible inference pipelines, automated evaluation, failure analysis, and comparative model benchmarking.
- Studied LLM orchestration, tool integration, Agentic AI, and Model Context Protocol (MCP) concepts for multi-step AI workflows involving retrieval, reasoning, validation, external tools, and response generation.
- Explored human-in-the-loop (HITL) and AI guardrail strategies for high-risk biomedical AI scenarios, focusing on uncertainty, response validation, explainability, and reliable decision support.
- Built PyTorch-based Variational Autoencoder (VAE) experiments on 780+ breast ultrasound images and evaluated latent representations using UMAP, t-SNE, clustering analysis, and downstream classification probes.
- Developed an uncertainty-aware diabetic-retinopathy grading framework using ResNet-18 and energy-based margin learning, analyzing confidence and ambiguous predictions for human-in-the-loop review.
- Built reproducible notebooks, experimental pipelines, automated evaluation workflows, visualizations, and technical documentation supporting model comparison, error analysis, research presentations, and peer-reviewed publications.

Environment: Python, PyTorch, Scikit-learn, Hugging Face Transformers, Sentence Transformers, Qwen, Mistral, LLMs, Generative AI, Retrieval-Augmented Generation (RAG), BGE Embeddings, ChromaDB, Vector Databases, Semantic Search, Information Retrieval, Prompt Engineering, Structured Outputs, LLM Evaluation, LLM Ops, Confidence Calibration, Grounding, Evidence Utilization, Hallucination Analysis, Response Validation, Agentic AI Concepts, Model Context Protocol (MCP) Concepts, Human-in-the-Loop (HITL), Trustworthy AI, Biomedical AI, Computer Vision, ResNet-18, Variational Autoencoders (VAE), UMAP, t-SNE, Git, Google Colab.

Hiten Technologies, INDIA
Junior AI/ML Engineer

January 2018 – November 2021

Responsibilities:

- Developed end-to-end machine learning and NLP solutions using Python, Scikit-learn, TensorFlow, and PyTorch for classification, prediction, document analysis, and business decision-support use cases.
- Built predictive models using Logistic Regression, Decision Trees, Random Forest, and XGBoost for risk scoring, forecasting, anomaly detection, and classification problems.
- Developed NLP pipelines for text preprocessing, tokenization, stemming, lemmatization, text classification, information extraction, and document analysis.
- Worked with transformer-based NLP models, including BERT, for text classification and contextual language-understanding use cases.
- Performed exploratory data analysis, missing-value treatment, outlier detection, feature engineering, model selection, and hyperparameter tuning to improve model performance.
- Evaluated models using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and regression metrics to support reliable model selection.
- Built scalable data preparation and feature-engineering pipelines using Python, SQL, Pandas, Spark/PySpark, and cloud-based data sources.
- Developed REST APIs using Flask/REST APIs to expose machine-learning predictions and integrate model outputs with downstream applications.
- Supported deployment and lifecycle management of ML models using Docker, MLflow, CI/CD practices, model versioning, and performance monitoring.
- Worked with AWS services including S3, Lambda, SageMaker, and CloudWatch to support data processing, model development, deployment, and monitoring.
- Built automated ETL and analytics workflows using Airflow and AWS data services, improving repeatability of data preparation and model-training processes.
- Collaborated with business stakeholders and technical teams to translate requirements into data-driven and machine-learning solutions, prototypes, dashboards, and production workflows.

Environment: Python, SQL, Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, PyTorch, Hugging Face/Transformers, BERT, NLP, Machine Learning, Flask, FastAPI, Apache Spark, PySpark, Airflow, MLflow, Docker, AWS S3, AWS Lambda, Amazon SageMaker, CloudWatch, PostgreSQL, Tableau, Git, CI/CD.

Independent / Freelance, INDIA
Data Analyst / Business Analyst

January 2013 – September 2015

Responsibilities:

- Analyzed operational and business datasets using Python, SQL, and Excel, applying statistical analysis, regression, forecasting, and predictive analytics to identify trends, performance gaps, and actionable business insights.
- Gathered and analyzed business requirements through stakeholder discussions, translating business needs into data requirements, functional specifications, KPIs, dashboards, reports, and analytical solutions.
- Worked with business and technical stakeholders to perform requirements analysis, process analysis, gap analysis, and data validation, helping align analytics solutions with business objectives and operational workflows.
- Designed ETL and data-processing workflows using Python and SQL to ingest, clean, transform, reconcile, and validate data from multiple source systems for analytics and reporting.
- Built interactive Tableau and Excel dashboards and executive reports to track KPIs, operational metrics, trends, and business performance, communicating analytical findings to technical and non-technical stakeholders.
- Supported UAT, data validation, defect/issue tracking, and requirements clarification, collaborating with stakeholders to verify that reports and analytics outputs met defined business requirements.
- Participated in Agile requirement discussions, sprint planning, documentation, and iterative delivery, maintaining business/technical documentation covering requirements, data sources, metric definitions, report logic, and validation procedures.

Environment: Python, SQL, Oracle, SQL Server, Advanced Excel, Excel Macros, Tableau, Data Analysis, Business Analysis, Requirements Gathering, Requirements Analysis, Functional Requirements, Process Analysis, Gap Analysis, UAT, Data Validation, Data Quality, Statistical Analysis, Predictive Analytics, Regression Analysis, Forecasting, ETL, KPI Reporting, Dashboard Development, Agile, Jira, Technical Documentation.